

- 7 (a) (i) path: reasonable curve upwards between plates B1
 straight and at a tangent to the curve beyond the plates B1 [2]
- (ii) 1. ($F =$) $E.g$ B1 [1]
2. ($t =$) L / v B1 [1]
- (b) (i) total momentum of a system remains constant **or** total momentum of a system before a collision equals total momentum after collision M1
 provided no external force acts on the system A1 [2]
 (do not accept 'conserved' but otherwise correct statement gets 1/2)
- (ii) ($\Delta p =$) EqL / v allow ecf from (a)(ii) B1 [1]
- (iii) *either* charged particle is not an isolated system M1
 so law does not apply A1 [2]
or system is particle and 'plates' (M1)
 equal and opposite Δp on plates / so law applies (A1)

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9 (c) (i) either $D = \sqrt{2} / D$ or $I = 1200 / 220$ or $E = 22$ C1